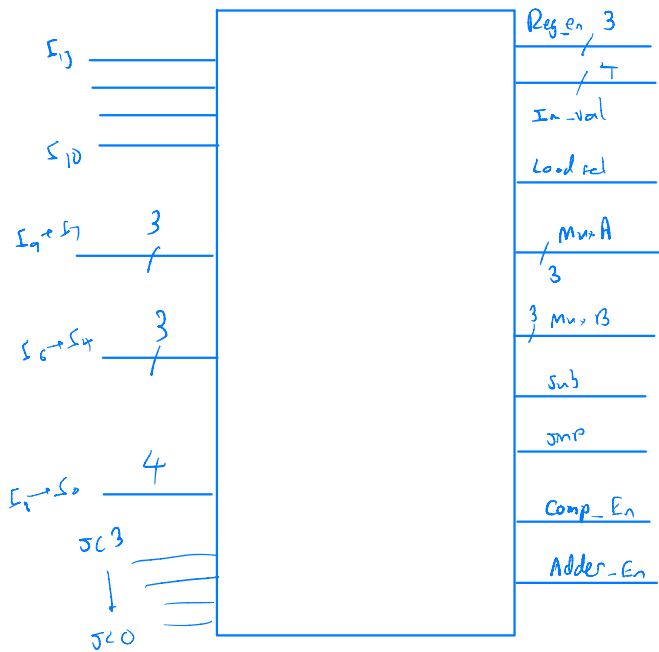




0000 $\rightarrow$ add
 0001 $\rightarrow$ sub
 0010 $\rightarrow$ movi
 0011 $\rightarrow$ jump
 0100 $\rightarrow$ snc
 0101 $\rightarrow$ neg
 0110 $\rightarrow$ dec
 0111 $\rightarrow$ reset
 1000 $\rightarrow$ com
 1001 $\rightarrow$ nop

Objectives

- Minimize Power consumption:
 - Adder will be on, only when necessary
 - Comp will be off until it's used
- No overwrites:
 - the register bank will not be overwritten

✓ inserted in analysis

		Add_En	comp_En	jmp	sub	Load_Sel	Reg_En	Sm_val	maxA, maxB
0 000	ADD ₁	1	0	0	0	0	$\overline{I_9} \cdot \overline{I_7}$	$a_0, 1, 15$	$a_0, 17, 15$
0 001	SUB ₂	1	0	0	1	0	$\overline{I_9} \cdot \overline{I_5}$		
0 010	MOV ₃	0	0	0	0	1	$\overline{I_9} \cdot \overline{I_7}$		
0 011	JER ₄	0	0	1 or 0	0	0	0000		
0 100	INC ₅	1	0	0	0	0	$\overline{I_9} \cdot \overline{I_7}$		
0 101	DEC ₆	1	0	0	0	0	$\overline{I_9} \cdot \overline{I_5}$		
0 110	NEG ₇	1	0	0	1	0	$\overline{I_9} \cdot \overline{I_5}$		
0 111	RES ₈	0	0	0	0	1	$\overline{I_9} \cdot \overline{I_5}$		
1 000	COM ₉	0	1	0	0	0	0000		
1 001	NOP ₁₀	0	0	0	0	0	X		
1 010		X	X	X	X	X	X		
1 011		X	X	X	X	X	X		
1 100		X	X	X	X	X	X		
1 101		X	X	X	X	X	X		
1 110		X	X	X	X	X	X		
1 111		X	X	X	X	X	X		

Add_En

$\overline{I_{11}} \cdot \overline{I_{10}}$ \ $\overline{I_{13}} \cdot \overline{I_{12}}$	00	01	11	10
00	1 ₁	1 ₂	0 ₃	0 ₄
01	1 ₅	1 ₆	0 ₇	1 ₈
11	X ₉	X ₁₀	X ₁₁	X ₁₅
10	0 ₁	0 ₁₀	X ₁₂	X ₁₁

$$Add_En = \overline{I_{13}} \cdot \overline{I_{11}} + \overline{I_{12}} \cdot \overline{I_{10}}$$

comp_En

$\overline{I_{11}} \cdot \overline{I_{10}}$ \ $\overline{I_{13}} \cdot \overline{I_{12}}$	00	01	11	10
00	0 ₁	0 ₂	0 ₃	0 ₄
01	0 ₅	0 ₆	0 ₇	0 ₈
11	X ₉	X ₁₀	X ₁₁	X ₁₅
10	1 ₁	0 ₁₀	X ₁₂	X ₁₁

$$comp_en = (\overline{I_{13}} + \overline{I_{11}} \cdot \overline{I_{10}})$$

$$= \overline{I_{12}} \cdot \overline{I_{11}} \cdot \overline{I_{10}}$$

jmp

$\overline{I_{11}} \cdot \overline{I_{10}}$ \ $\overline{I_{13}} \cdot \overline{I_{12}}$	00	01	11	10
00	0 ₁	0 ₂	1 ₃	0 ₄
01	0 ₅	0 ₆	0 ₇	0 ₈
11	X ₉	X ₁₀	X ₁₁	X ₁₅
10	0 ₁	0 ₁₀	X ₁₂	X ₁₁

$$jmp = \overline{I_{11}} \cdot \overline{I_{11}} \cdot \overline{I_{10}}$$

Sub

$I_{11} I_{10}$	00	01	11	10
$I_{13} I_{12}$				
00	0 ₁	1 ₂	0 ₃	0 ₄
01	0 ₅	0 ₆	0 ₇	1 ₈
11	X ₉	X ₁₀	X ₁₁	X ₁₂
10	0 ₁	0 ₁₀	X ₁₂	X ₁₁

$$Sub = \bar{I}_{13} \cdot \bar{I}_{11} \cdot I_{10} + I_{12} \cdot I_{11} \cdot \bar{I}_{10} \cdot \bar{I}_{12}$$

Load-SEL

$I_{11} I_{10}$	00	01	11	10
$I_{13} I_{12}$				
00	0 ₁	0 ₂	0 ₃	1 ₄
01	0 ₅	0 ₆	1 ₇	0 ₈
11	X ₉	X ₁₀	X ₁₁	X ₁₂
10	0 ₁	0 ₁₀	X ₁₂	X ₁₁

$$Load-SEL = I_{12} \cdot I_{11} \cdot I_{10} + \bar{I}_{12} \cdot I_{11} \cdot \bar{I}_{10} \\ = I_{11} \cdot (I_{12} I_{10} + \bar{I}_{12} \bar{I}_{10}) \\ = I_{11} \cdot (I_{12} \oplus \bar{I}_{10})$$

Reg-SEL

$I_{11} I_{10}$	00	01	11	10
$I_{13} I_{12}$				
00	1 ₁	1 ₂	0 ₃	1 ₄
01	1 ₅	1 ₆	1 ₇	1 ₈
11	X ₉	X ₁₀	X ₁₁	X ₁₂
10	0 ₁	X ₁₁	X ₁₂	X ₁₁

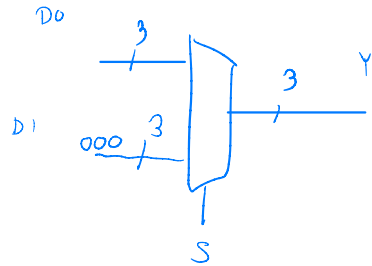
$$Reg-SEL = I_{13} + \bar{I}_{12} I_{11} I_{10}$$

* connect to a 2:1 3-bit

max

$$0 \rightarrow I_9 - I_7$$

$$1 \rightarrow 0000$$



Instruction set

OP code	func	Syntax	Details	Format
0000	ADD	ADD R _a , R _b	Add R _a + R _b & overwrite R _a	0000 R _a R _a R _a R _b R _b R _b 0000
0001	SUB	SUB R _a , R _b	Sub R _b - R _a & overwrite R _a	0001 R _a R _a R _a R _b R _b R _b 0000
0010	MOV	MOV R _j , d	Move d to R	0010 RRR 000 dddd
0011	JZR	JZR R, d	Jump to line d if R = 0	0011 RRR 000 oddd
0100	INC	INC R	Increase R by 1	0100 RRR 001 0000
0101	DEC	DEC R	Decrease R by 1	0101 RRR 010 0000
0110	NEG	NEG R	Negate R	0110 RRR 000 0000
0111	RES	RES R	Reset R to 0	0111 RRR 000 0000
1000	COM	COM R _a , R _b	Compare R _a & R _b	1000 R _a R _a R _a R _b R _b R _b 0000
1001	NOP	NOP	Do nothing	1001 000 000 0000